6 Multistage Amplifiers

INTRODUCTION

- In the previous few chapters, we have discussed about the AC and DC analysis of a single amplifier. But in the real world and in the electronic circuitry, we don't have a single amplifier but a group of cascaded amplifiers with a variable input source.
- Sometimes, in a single amplifier, we apply input signals of different frequencies because the signals which we want to amplify are of different frequencies. so, resistor and capacitor components connected in the circuit change it's behaviour due to the variation in the input voltage.
- In this chapter, we will discuss about the behaviour of an amplifier due to variation in frequency and loading effect.
- Loading effect refers to the effect of the second stage amplifier on the first stage. The combination of two or more amplifiers in a single electronic circuit is known as multistage amplifiers.
- We will study the frequency response of BJT and multistage amplifiers followed by FET.

Frequency response of BJT:

- So far, we have assumed that frequency is high enough so that coupling capacitors and bypass capacitors get shorted and at the same time frequency is low enough that parasitic load & transistor capacitance are open.
- Frequency response allows us to vary the frequency of input signal & observe the variation of gain.

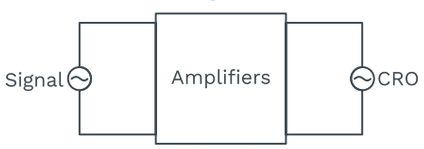


Fig. 6.1 Amplifier Circuit with CRO

Rack Your Brain

Why a single amplifier is not able to provide much larger gain?

Grey Matter Alert!

With a single amplifier used at different times, we apply different signals for amplification which is obviously different frequencies.

Rack Your Brain

Why behaviour of different capacitors is different in a different range of frequencies?

- V_i = constant voltage peak to peak then f is varied from Hz to MHz and observed at all frequencies.

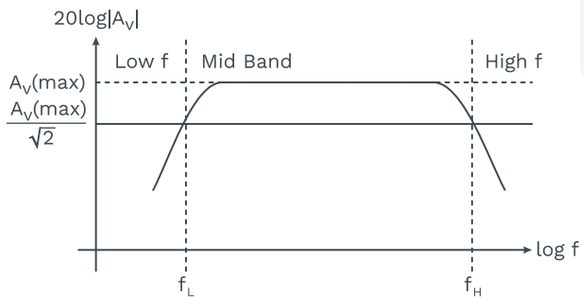


Fig. 6.2 Frequency Response of BJT

f_L = lower cut-off frequency
f_H = upper cut-off frequency
Till now, we have discussed about the mid band range of frequency. In this chapter we will discuss about the low frequency and high frequency range of the amplifier.
The behaviour of the capacitors are changed according to the frequency.
Coupling capacitor C_{C1} and C_{C2}
Diffusion capacitor C_D
Emitter bypass capacitor C_E
Transition capacitor C_T
Load capacitor C_L
All the capacitors listed above are shown below.

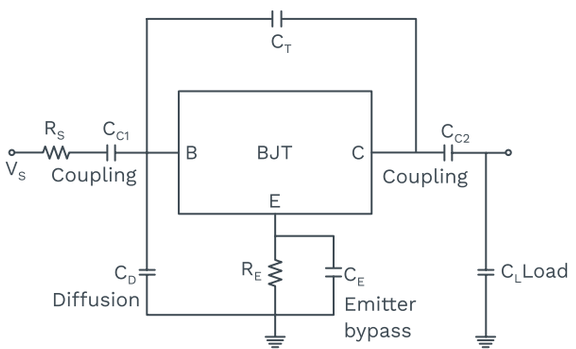


Fig. 6.3 BJT Amplifiers with all Capacitors

Rack Your Brain

What is the corner frequency?

Previous Years' Question

The current gain of a bipolar transistor drops at high frequencies because of:
A) transistor capacitances
B) high current effects in the base
C) parasitic inductive elements
D) the early effect

Grey Matter Alert!

Transition capacitance (C_T) and diffusion capacitance (C_D) are the internal capacitances of transistors formed between different layers of the transistor.

Frequency Range	Coupling and Bypass Capacitor (μF)	Load And Parasitic Capacitor (in nF OR pF)
Low frequency	Consider	Open
Mid band	Short	Open
High frequency	Short	Consider

Table 6.1

1) Low frequency response:

Low frequency response of a BJT is similar to a high pass filter.

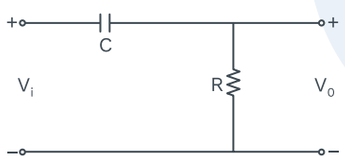


Fig. 6.4 Low Frequency Response

$$\frac{V_0}{V_i} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega RC}{1 + j\omega RC}$$

$$= \frac{1}{1 - \frac{j}{\omega RC}} = \frac{1}{1 - j\left(\frac{\omega_L}{\omega}\right)}$$

$$\omega_L = \frac{1}{RC}$$

$$\left|\frac{V_0}{V_i}\right| = \frac{1}{\left[1 + \left(\frac{\omega_L}{\omega}\right)^2\right]^{1/2}}$$

$$\phi = \tan^{-1}\left(\frac{\omega_L}{\omega}\right)$$

Since $\phi > 0$, this network can work as a lead compensator.

$$\left|\frac{V_0}{V_i}\right| = \frac{1}{\left(1 + \left(\frac{f_L}{f}\right)^2\right)^{1/2}}$$

Grey Matter Alert!

3dB bandwidth is also known as half power bandwidth.

Previous Years' Question

Generally, the gain of a transistor amplifier falls at high frequencies due to the:

a) internal capacitances of the device
b) coupling capacitor at the input
c) skin effect
d) coupling capacitor at the output

at $f \to \infty$, $\left|\frac{V_0}{V_i}\right| = 1 \to$ maximum gain

at $f = f_L$, $\left|\frac{V_0}{V_i}\right| = \frac{1}{\sqrt{2}} = \frac{\text{maximum}}{\sqrt{2}}$

f_L = cut off frequency $= \frac{1}{2\pi RC}$ Hz

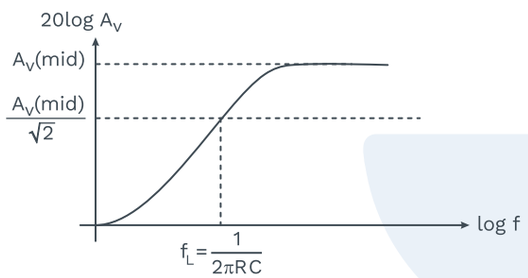


Fig. 6.5 Response of High Pass Filter

In low frequency response, we have to determine resistance across the capacitor and then the cutoff frequency $= \frac{1}{2\pi RC}$ Hz

Low frequency response of common emitter amplifier:

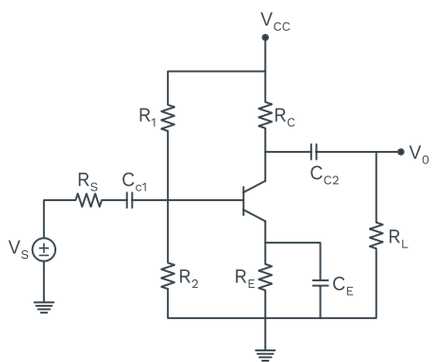


Fig. 6.6 Common Emitter Amplifier

- At low frequency C_{C1}, C_{C2} and C_E cannot be shorted & they must be considered.

Rack Your Brain

How capacitor C_E affects the output impedance of an amplifier circuit?

Grey Matter Alert!

Bandwidth is the range of frequencies over which gain is greater than 3dB.

Rack Your Brain

What is the unity gain bandwidth?

AC model:

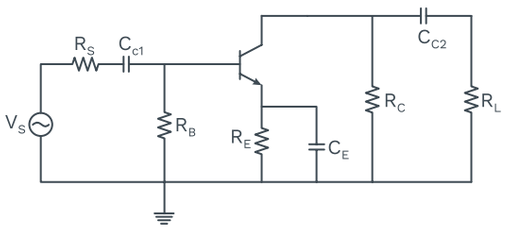


Fig. 6.7 CE AC Model

$$R_B = R_1 \parallel R_2$$

- We will compute three cut-off frequencies, one each corresponding to C_{C1}, C_{C2} & C_E.

For this purpose, we need to compute resistance across each of these capacitors.

Effect of C_{C1}:

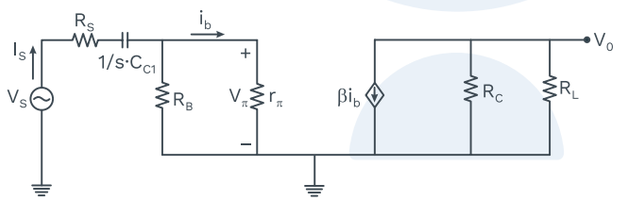


Fig. 6.8 Effect of C_{C1}

$$V_\pi = \frac{V_S[R_1 \parallel R_2 \parallel r_\pi]}{R_S + \frac{1}{sC_{C1}} + R_1 \parallel R_2 \parallel r_\pi}$$

$$i_b = \frac{V_\pi}{r_\pi}$$

$$V_0 = -\beta i_b (R_C \| R_L)$$

$$V_0 = \frac{-\beta V_\pi (R_C \parallel R_L)}{r_\pi}$$

$$\frac{V_0}{V_S} = -\frac{\beta (R_C \| R_L)(R_1 \| R_2 \| r_\pi)}{r_\pi \left[R_s + (R_1 \parallel R_2 \parallel r_\pi) + \frac{1}{sC_{C1}} \right]}$$

Taking $R_s+(R_1\|R_2\|r_\pi)$ common in denominator, we get:

Grey Matter Alert!

Bypass capacitors play a very significant role in the gain and input impedance of the amplifier.

$$\text{Gain} = -\frac{\beta(R_C \| R_L)(R_1 \| R_2 \| r_\pi)}{r_\pi\left[R_s + \left(R_1 \| R_2 \| r_\pi\right)\right] \times \left(1 + \frac{1}{sC_{C1}\left[R_S + R_1 \| R_2 \| r_\pi\right]}\right)}$$

$$\text{Cut off frequency} = \frac{1}{2\pi[R_S + (R_1 \| R_2 \| r_\pi)]C_{C1}}$$

Effect of bypass capacitor (C_E):

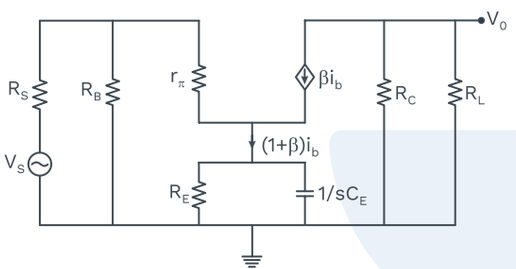


Fig. 6.9 Effect of Bypass Capacitor (C_E)

- To consider the effect of C_E, short circuit C_{C1} and C_{C2}.
- Short circuit V_S and connect a new source in place of C_E.
- We have ignored R_s to simplify the calculation

Grey Matter Alert!

Break frequency is also known as the corner frequency.

Rack Your Brain

What is the upper cutoff frequency?

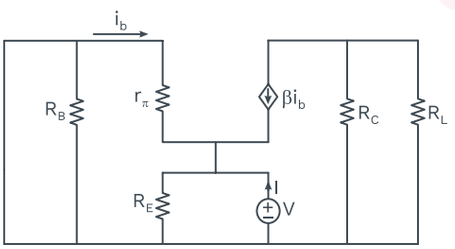


Fig. 6.10 Simplified Effect of C_E

$$V = (I + i_b + \beta i_b)R_E$$

$$i_b r_\pi + V = 0$$

$$i_b = \frac{-V}{r_\pi}$$

$$V = IR_E + (1+\beta)R_E\left(\frac{-V}{r_\pi}\right)$$

$$V\left[1+\frac{(1+\beta)R_E}{r_\pi}\right]=IR_E$$

$$\frac{V}{I}=\frac{R_E}{1+\frac{(1+\beta)R_E}{r_\pi}}=R_{eq}$$

$$f_{L2}=\frac{1}{2\pi R_{eq}C_E}$$

Grey Matter Alert!

Amplitude and frequency distortion in an amplifier are mainly due to the non-linear nature of BJT and FET.

Effect of C_{C2}:

Output side circuit

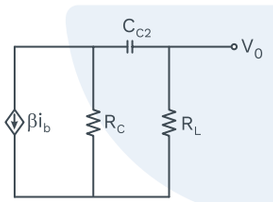


Fig. 6.11 Effect of C_{C2}



Fig. 6.12 Equivalent Resistances Across C_{C2}

Equivalent resistance = R_C+R_L

$$f_{L3}=\frac{1}{2\pi(R_C+R_L)C_{C2}}$$

Unity gain frequency:

- The frequency at which the gain of the amplifier becomes unity is called unity gain frequency.
- In high pass filter (HPF), low frequency response of BJT is given as:

$$\text{Gain}=\frac{k}{\left[1+\left(\frac{\omega_L}{\omega}\right)^2\right]^{1/2}}$$

at $\omega = \omega_T$; gain = 1

(ω_T: Unity gain frequency)

Previous Years' Question

The 3-dB bandwidth of the low pass signal $e^{-t}u(t)$, where u(t) is the unit step function, is given by:

a) $\frac{1}{2\pi}$ Hz **b)** $\frac{1}{2\pi}\sqrt{\sqrt{2}-1}$ Hz

C) ∞ **d)** 1 Hz

$$1 = \frac{k}{\left(1+\left(\frac{\omega_L}{\omega_T}\right)^2\right)^{1/2}}$$

$\omega_T < \omega_L$ then $\frac{\omega_L}{\omega_T} >> 1$

$$1 = \frac{k}{\left(\frac{\omega_L}{\omega_T}\right)}$$

$$\Rightarrow \omega_T = \frac{\omega_L}{k}$$

Computing overall f_L:

- Lower cut-off frequency of the system is a maximum of three lower cut-off frequencies.

$$f_L = \max(f_{L1}, f_{L2}, f_{L3})$$

2) High frequency response:

The high frequency response of BJT is similar to a low pass filter.

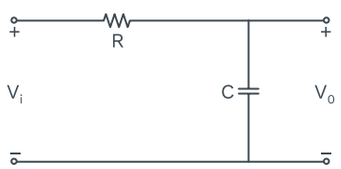


Fig. 6.13 High Frequency Response of BJT

$$\frac{V_0}{V_i} = \frac{1/j\omega C}{R + \frac{1}{j\omega C}}$$

$$\frac{V_0}{V_i} = \frac{1}{1+j\omega RC} = \frac{1}{1+j(\omega/\omega_H)}$$

$$\omega_H = \frac{1}{RC}$$

$$\left|\frac{V_0}{V_i}\right| = \frac{1}{\left[1+\left(\frac{\omega}{\omega_H}\right)^2\right]^{1/2}} = \frac{1}{\left[1+\left(\frac{f}{f_H}\right)^2\right]^{1/2}}$$

Rack Your Brain

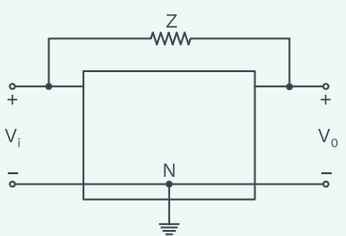
Why Miller's effect is very important in frequency response of an amplifier?

Grey Matter Alert!

Miller's effect gives two impedances from a single impedance.

Previous Years' Question

The circuit shown in the figure is a finite gain amplifier with a gain of K, a very large input impedance, and a very low output impedance. The input impedance of the feedback amplifier with the feedback impedance Z connected as shown will be:



a) $Z\left[1-\frac{1}{K}\right]$ **b)** $Z(1-K)$

c) $\left[\frac{Z}{K-1}\right]$ **d)** $\left[\frac{Z}{1-K}\right]$

f_H = upper cutoff frequency $= \frac{1}{2\pi RC}$

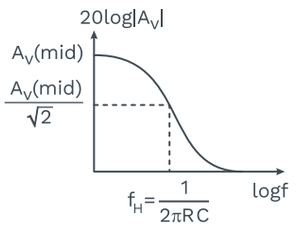


Fig. 6.14 Bode Plot

$$\phi = -\tan^{-1}\left(\frac{f}{f_H}\right)$$

Since, $\phi < 0$, it can work as a lag compensator

$$\text{as } f = 0, \left|\frac{V_0}{V_i}\right| = 1$$

$$\text{as } f = f_H, \left|\frac{V_0}{V_i}\right| = \frac{1}{\sqrt{2}}$$

Miller's theorem:

- To simplify the network when an impedance is connected between the two ports we can apply Miller's theorem to replace the impedance between the ports by two impedances one at each port.

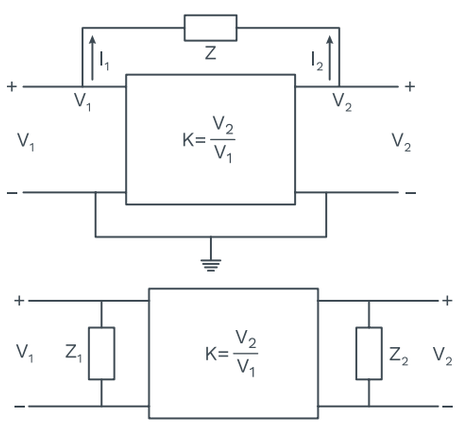


Fig. 6.15 Miller's Effect

Rack Your Brain

What is the Giacoletto model and how is it useful?

Grey Matter Alert!

If the capacitance used in the amplifier is large; bandwidth would be small and vice-versa.

Previous Years' Question

The f_T of a BJT is related to its g_m, C_π and C_μ as follows:

a) $f_T = \frac{C_\pi + C_\mu}{g_m}$

b) $f_T = \frac{2\pi(C_\pi + C_\mu)}{g_m}$

c) $f_T = \frac{g_m}{C_\pi + C_\mu}$

d) $f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$

$$z_1 = \frac{z}{1-k} \qquad z_2 = \frac{kz}{k-1}$$

$$I_1 = \frac{V_1 - V_2}{z} = \frac{V_1 - kV_1}{z} = \frac{V_1}{z/(1-k)}$$

From the 2nd circuit,

$$I_1 = \frac{V_1}{z_1} \text{ where } z_1 = \frac{z}{(1-k)}$$

$$I_2 = \frac{V_2 - V_1}{z} = \frac{V_2 - V_2 / k}{z} \text{ (from the 1st circuit)}$$

From the 2nd circuit,

$$I_2 = V_2/Z_2 \text{ where, } z_2 = \frac{z}{(1-1/k)}$$

Hybrid π-model at high frequency:

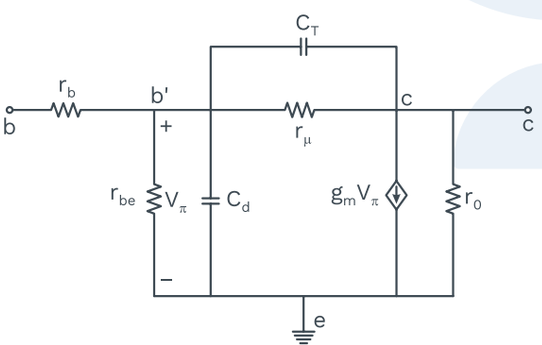


Fig. 6.16 Hybrid π-model at High Frequency

r_b = base spreading resistance

$r_{be} = r_\pi = \beta r_e$

r_μ = due to early effect (base to collector resistance or feedback resistance)

r_0 = due to early effect (output resistance or collector to emitter resistance)

- Increase in reverse bias across CB junction reduces base width & increases I_E, I_C & reduces I_B, so output voltage impacts input current and is represented by r_μ.

Rack Your Brain

Why current gain of common emitter configuration is very high?

Grey Matter Alert!

Square wave input can be used to determine the higher and lower cutoff frequencies.

Previous Years' Question

An npn transistor has a beta cut-off frequency f_β of 1MHz and common emitter short circuit low frequency current gain b_0 of 200. Its unity gain frequency f_T and the alpha cut-off frequency f_α respectively are:

a) 200 MHz, 201 MHz
b) 200 MHz, 199 MHz
c) 199 MHz, 200 MHz
d) 201 MHz, 200 MHz

Hybrid model:

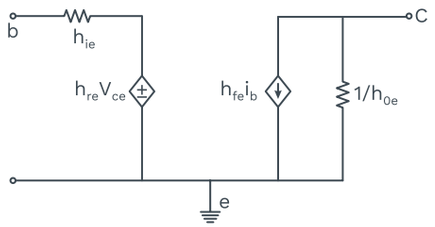


Fig. 6.17 Hybrid Model

- Relation with hybrid π model,

Transconductance $g_m = \frac{I_{CQ}}{V_T}$

Input resistance $r_{b'e} = r_\pi = \beta r_e$

$$= \frac{\beta V_T}{I_{EQ}} \approx \frac{\beta V_T}{I_{CQ}}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{h_{fe}}{g_m}$$

Feedback resistance,

$$r_\mu = r_\pi \frac{(1 - h_{re})}{h_{fe}}$$

Base spreading resistance,

$$r_{bb'} = h_{ie} - r_\pi$$

Output resistance $r_0 = \frac{1}{h_{oe} - \frac{(1 + h_{fe})}{r_\mu}}$

Giacoletto model by ignoring capacitors

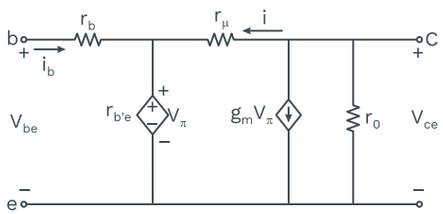


Fig. 6.18 Giacoletto Model

$V_{be} = i_b r_b + V_\pi$

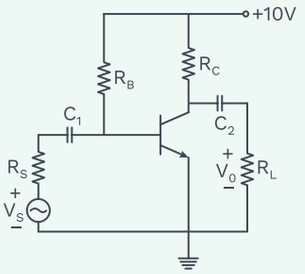
Rack Your Brain

What is α-cutoff frequency and its relation with transconductance g_m?

Previous Years' Question

Consider the common emitter amplifier shown below with the following circuit parameters:

$\beta = 100$, $g_m = 0.3861 A/V$, $r_0 = \infty$
$r_\pi = 259\Omega$, $R_s = 1k\Omega$, $r_B = 93k\Omega$,
$R_C = 250\Omega$, $R_L = 1k\Omega$, $C_1 = \infty$
and $C_2 = 4.7\mu F$



- The resistance seen by the source V_s is:
  a) 258Ω **b)** 1258Ω
  c) 93kΩ **d)** ∞
- The lower cut-off frequency due to C_2 is:
  a) 33.9 Hz **b)** 27.1 Hz
  c) 13.6 Hz **d)** 16.9 Hz

By nodal analysis;

$$\frac{V_\pi}{r_{b'e}} + \frac{V_\pi - V_{ce}}{r_\mu} + \frac{V_\pi - V_{be}}{r_b} = 0$$

$$V_\pi\left[\frac{1}{r_{b'e}} + \frac{1}{r_\mu} + \frac{1}{r_b}\right] = \frac{V_{ce}}{r_\mu} + \frac{V_{be}}{r_b}$$

$$V_\pi = \frac{\frac{V_{ce}}{r_\mu} + \frac{V_{be}}{r_b}}{\frac{1}{r_{b'e}} + \frac{1}{r_\mu} + \frac{1}{r_b}}$$

$$V_{be} = i_b r_b + V_\pi$$

$$V_{be} = i_b r_b + \frac{V_{ce}/r_\mu + V_{be}/r_b}{\left(\frac{1}{r_{b'e}} + \frac{1}{r_\mu} + \frac{1}{r_b}\right)}$$

$$V_{be}\left[1 - \frac{1/r_b}{\frac{1}{r_{b'e}} + \frac{1}{r_\mu} + \frac{1}{r_b}}\right] = i_b r_b + \frac{V_{ce}/r_\mu}{\left(1/r_{b'e} + 1/r_\mu + 1/r_b\right)}$$

From h-model, we get:

$$V_{be} = i_b h_{ie} + h_{re} V_{ce}$$

By comparing, we derive the relation between two parameters.

Collector emitter short circuit current gain:

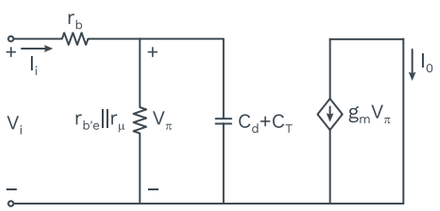


Fig. 6.19 CE Current Gain

When CE terminals are shorted the resistance and capacitor between base and collector get connected between base and emitter.

Grey Matter Alert!

1 dB is the same as 1 octave.

Rack Your Brain

What is the difference between the interacting and non-interacting stages?

Grey Matter Alert!

The loading effect for a cascaded amplifier must be as low as possible.

$$I_0 = -g_m V_\pi$$

$$I_i = Y_i V_i = \frac{V_i}{Z_i}$$

Current gain

$$\frac{I_0}{I_i} = -g_m Z_i \left(\frac{V_\pi}{V_i}\right)$$

$$\frac{V_\pi}{V_i} = \frac{\left[r_{b'e} \parallel r_\mu \parallel \frac{1}{sC}\right]}{r_b + \left(r_{be} \parallel r_\mu \parallel \frac{1}{sC}\right)}$$

$$C = C_d + C_T$$

Current gain

$$= -g_m \left[r_b + \left(r_{b'e} \parallel r_\mu \parallel \frac{1}{sC}\right)\right] \times \frac{\left[r_{b'e} \parallel r_\mu \parallel \frac{1}{sC}\right]}{r_b + \left(r_{b'e} \parallel r_\mu \parallel \frac{1}{sC}\right)}$$

$$= -g_m \left(r_{b'e} \parallel r_\mu \parallel \frac{1}{sC}\right)$$

$$\approx \frac{-g_m \times r_{b'e}}{1 + sC\, r_{b'e}} = \frac{-g_m r_\pi}{1 + sCr_\pi}$$

$$r_\pi = \frac{\beta}{g_m}$$

$$\therefore \; A_I = \frac{-\beta}{1 + sCr_\pi} = \frac{-\beta}{\left(1 + \frac{j\omega}{\omega_H}\right)}$$

- Cut-off frequency for short circuit conditions is:

$$f_c = \frac{1}{2\pi r_\pi C}, C = C_d + C_T$$

Unity gain frequency:

Current gain, $A_I = \dfrac{-\beta}{1 + \frac{j\omega}{\omega_H}}$

$$|A_I| = \frac{\beta}{\left[1 + \left(\frac{\omega}{\omega_H}\right)^2\right]^{1/2}}$$



Rack Your Brain

Why bandwidth of an amplifier decreases in a multistage amplifier?

Grey Matter Alert!

The product of gain and bandwidth of an amplifier is always constant.

at $\omega = \omega_T$, $|A_I| = 1$

$$1 = \frac{\beta}{(\omega_T/\omega_H)}; \left[\frac{\omega_T}{\omega_H} >> 1\right]$$

$$\omega_T = \beta\omega_H = \frac{\beta}{r_\pi C}$$

At $\omega \to 0$, $A_I = -\beta$

- gain × BW = constant
- at unity gain frequency ω_T; gain = 1

$$\omega_T = \text{gain} \times \omega_{3dB}$$

$$1 \times \omega_T = \beta \times \omega_H$$

To be remembered, $\omega_T = \frac{\beta}{r_\pi C} = \frac{g_m}{C}$

- α-cut-off frequency (f_α)

$1 \cdot f_T = \alpha \cdot f_\alpha$

$f_\alpha = \frac{f_T}{\alpha}$; $f_T = \frac{\omega_T}{2\pi} = \frac{g_m}{2\pi C}$

- β-cut-off frequency (f_β)

$1 \cdot f_T = \beta f_\beta$

$f_\beta = \frac{f_T}{\beta}$

as gain ↑, BW↓

Multistage amplifier:

Why should we need a multistage amplifier? or what are the disadvantages of single, stage amplifier?

There are some drawbacks of single stage amplifiers:

- Gain is less.
- Input and output resistance requirements cannot be met.
- Poor efficiency.

Previous Years' Question

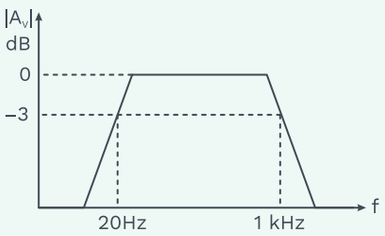
An amplifier has an open-loop gain of 100 and its lower and upper-cut-off frequencies are 100 Hz and 100 KHz respectively. A feedback network with a feedback factor of 0.99 is connected to the amplifier. The new lower and upper-cut-off frequencies are at _____ Hz and _____ MHz.

Rack Your Brain

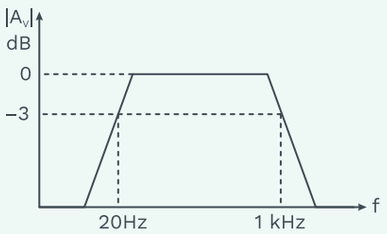
What are the applications of an RC coupled transformers?

Previous Years' Question

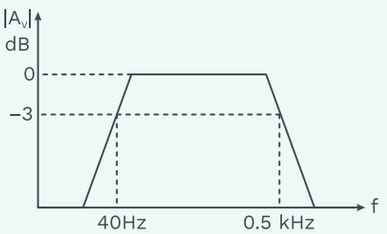
Three identical RC-coupled transistor amplifiers are cascaded. If each of the amplifiers has a frequency response as shown in the figure, the overall frequency response:



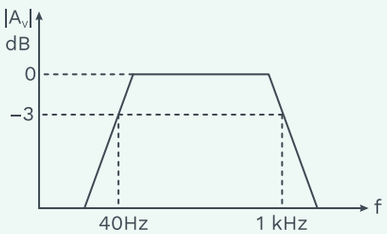
A)



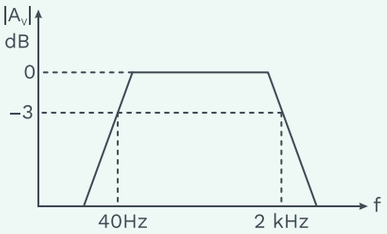
B)



C)



D)



Cascading of amplifiers:

By cascading amplifiers, the gain increases but since gain × BW product is constant the BW reduces.

This limitation is overcome by a cascade amplifier, which is a large gain and large BW amplifier.

1) Cascading of non-interacting stages

Non-interacting stages imply that there is no current between the two stages due to which there is no voltage drop, or we can say there is the loading effect.

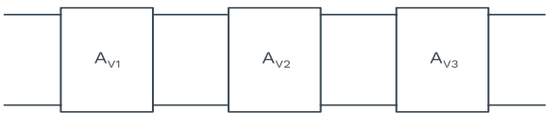


Fig. 6.20 Cascade Amplifier

Loading effect: When the second stage draws a current from 1st stage of the amplifier, then it acts like a load on 1st stage and hence called as loading effect.

$$A_{V1} = A_{V2} = A_{V3} = A_V \text{ (Identical stages)}$$

$$f_{L1} = f_{L2} = f_{L3} = f_L$$
$$f_{H1} = f_{H2} = f_{H3} = f_H$$

Overall gain = $(A_{V1} \cdot A_{V2} \cdot A_{V3}) = (A_V)^3$

If n-such stages are cascaded, then the overall gain = $(A_V)^n$

Lower cutoff frequency of cascaded amplifier = F_L

In low frequency region, the response is similar to the high pass filter (HPF) response.

Gain for single stage amplifier is given by,

$$\text{Gain} = \frac{A_{V(mid)}}{\sqrt{1+\left(\frac{f_L}{f}\right)^2}}$$

For n cascaded stage, gain will be,

$$\text{Gain} = \left[\frac{A_{V(mid)}}{\left(1+\left(\frac{f_L}{f}\right)^2\right)^{1/2}}\right]^n$$

On comparing, we get:

$$A_{mid} = [A_{V(mid)}]^n$$

$$1+\left(\frac{f_L}{f}\right)^2 = \left[1+\left(\frac{f_L}{f}\right)^2\right]^n$$

$$F_L = \frac{f_L}{\left(2^{1/n}-1\right)^{1/2}}$$

- Similarly, for high frequency regions,

$$F_H = f_H(2^{1/n}-1)^{1/2}$$

Rack Your Brain

How two common collector configuration amplifiers increase the current gain?

Grey Matter Alert!

Bandwidth of the direct coupled amplifier is maximum.

Where F_H is a higher cutoff frequency of n cascaded stage.

and f_H is a higher cutoff frequency of single stage.

- The bandwidth of the cascaded amplifier
$$= F_H - F_L \approx F_H$$
- From the above formula mentioned above, $F_H < f_H$

∴ BW reduces after n cascaded stage.

- Bandwidth is the range of frequencies over which the gain of the amplifier almost remains constant during mid-band frequencies.

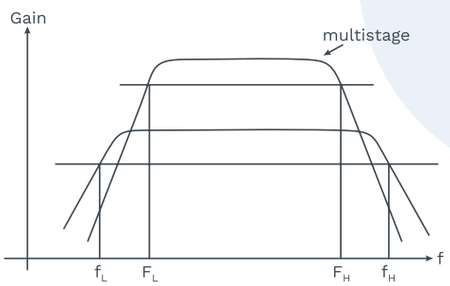


Fig. 6.21 Gain vs Frequency

Cascading interacting stage:

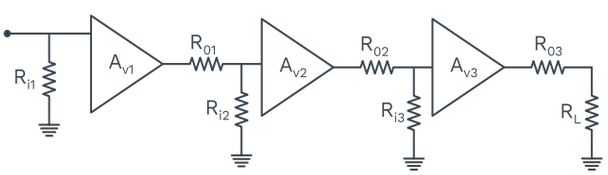


Fig. 6.22 Cascade Interacting Stage

- Now, we will consider that stages have finite input resistance due to which current flows between stages, and there is a loading effect.

Overall gain:

$$A_1 = \frac{A_{v1} \cdot R_{i2}}{R_{01} + R_{i2}}$$

$$A_2 = A_{v2} \frac{R_{i3}}{R_{02} + R_{i3}}$$

Previous Years' Question

An RC-Coupled Amplifier is assumed to have a single-pole low frequency transfer function. The maximum lower-cut-off frequency allowed for the amplifier to pass 50 Hz square wave with no more than 10% tilt is _____ Hz.

Rack Your Brain

Why voltage gain of the Darlington pair amplifier is very poor?

Grey Matter Alert!

The main drawback of the Darlington pair amplifier is reverse saturation of the first stage, which is temperature sensitive and gets amplified.

$$A_3 = A_{v3}\frac{R_L}{R_L + R_{03}}$$

Overall gain = $A_1A_2A_3$

The gain is smaller as compared to gain of non-interacting stages.

Overall cut-off frequencies:

$F_L > \max(f_{L1}, f_{L2}, f_{L3})$

& $F_H < \min(f_{H1}, f_{H2}, f_{H3})$

$F_L = [f_{L1}^2 + f_{L2}^2 + f_{L3}^2]^{1/2}$

$$\frac{1}{F_H} = \left[\frac{1}{f_{H1}^2} + \frac{1}{f_{H2}^2} + \frac{1}{f_{H3}^2}\right]^{1/2}$$

- **Rise time:**

$t_r = [t_{r1}^2 + t_{r2}^2 + t_{r3}^2]^{1/2}$

$$t_r \uparrow, BW \downarrow \Rightarrow BW \propto \frac{1}{t_r}$$

Type of amplifiers coupling:

1) RC coupling:

- The purpose of coupling capacitor in a single stage amplifier is to connect the source & load to the amplifier.
- The two stages of the amplifier are coupled through a capacitor and resistor which is in the input resistance of the second stage.

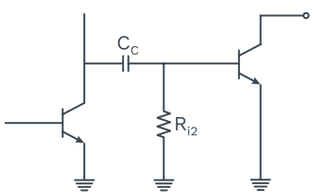


Fig. 6.23 RC Coupling

Advantage:

- It offers a constant gain over a wide frequency band.
- It is low cost because resistors and capacitors are not expensive.
- The circuit is very compact and light.

Disadvantage:

It has poor low frequency and high frequency gain. At low frequency, gain is reduced due to coupling capacitor, and at high frequency gain is reduced due to load and parasitic capacitance.

2) Direct coupled amplifier:

There is no coupling capacitor between the two stages of the amplifier.

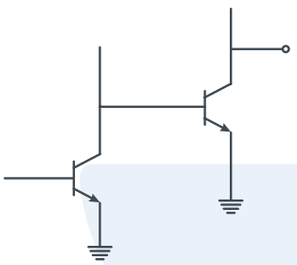


Fig. 6.24 Direct Coupled Amplifier

Advantage:

- It can amplify both AC and DC due to the absence coupling capacitor, so it has a better low frequency response and is hence used as an industrial instrumentation amplifier.
- It has a large gain.

Disadvantage:

- The Q point is unstable because the stages are interacting, and it is called as dc drift.
- Due to dc drifts, the output will be distorted.

CC-CC amplifiers or Darlington pair:

- If the two transistors are cascaded in a common collector configuration, then it is called as Darlington Pair.
- For common collector pair

$$h_{ic} = h_{ie}$$

$$h_{fc} = 1 + h_{fe}$$

The approximate hybrid model can be used if $h_{oe}R_L < 0.1$ in which we ignore h_0.

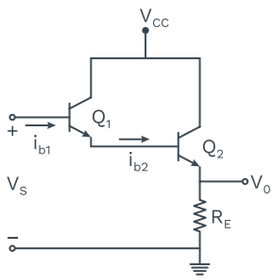


Fig. 6.25 Darlington Pair Amplifier

- In both transistors, input is applied at the base, and output is taken from the emitter & hence it is common for a collector.
- Analyzing 2nd stage $h_{oe}R_E < 0.1$. So we use an approximate hybrid model.

$$A_{I2} = \frac{I_0}{I_{b2}} = 1 + h_{fe}$$

$$R_{i2} = (1+h_{fe})R_E$$

Voltage gain:

$$A_{V2} = \frac{V_0}{V_{i2}} = \frac{I_0 R_E}{I_{b2} R_{i2}} = A_{I2} \times \frac{R_E}{(1+h_{fe})R_E} = 1$$

For 1st stage

The input resistance of the second stage acts as load on 1st stage $R_{L1}=(1+h_{fe})R_E$

Now, $h_0R_L \leq = 1$, so it is no longer valid. So we have to use the exact hybrid model.

Current gain:

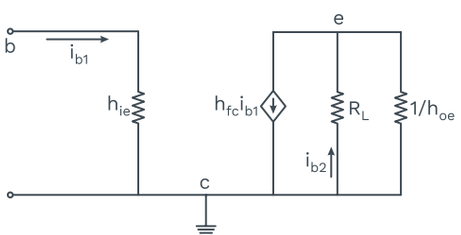


Fig. 6.26 Current Gain of Darlington Pair

$$A_{I1} = \frac{i_{b2}}{i_{b1}} = \frac{h_{fc} \times i_{b1} \times \frac{1/h_{oe}}{R_L + 1/h_{oe}}}{i_{b1}} = \frac{h_{fc}}{1 + h_{oe}R_L}$$

$$A_{i1} = \frac{1+h_{fe}}{1+h_{oe}R_L} \qquad (h_{fc} = 1+h_{fe})$$

$$= \frac{(1+h_{fe})}{1+h_{oe}(1+h_{fe})R_E} \qquad (R_L = (1+h_{fe})R_E)$$

$$A_{i1} = \frac{1+h_{fe}}{1+h_{oe}h_{fe}R_E} \qquad (1+h_{fe} \approx h_{fe})$$

Input resistance:

$R_{i1} = h_{ie} + h_{rc} A_{i1} R_{i2}$

$= h_{ie} + A_{i1} R_{i2} \qquad \{h_{rc} = 1\}$

From current gain expression:

$$R_{i1} = h_{ie} + \left(\frac{1+h_{fe}}{1+h_{oe}h_{fe}R_E}\right) \times (1+h_{fe})R_E$$

h_{ie} is very small, so neglect it.

$$\therefore R_{i1} = \frac{(1+h_{fe})^2}{1+h_{oe}h_{fe}R_E} \cdot R_E$$

Rack Your Brain

How the frequency response of MOSFET is different from that of BJT?

Grey Matter Alert!

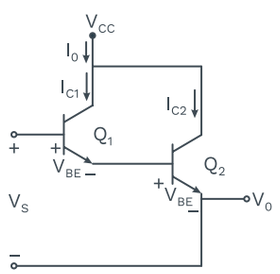
In an op-amp circuit, the gain bandwidth product is the same as unity gain bandwidth.

Important points regarding Darlington pair:

- Both transistors are in a common collector configuration.
- For the 2nd stage, we use an approximate hybrid model.
- For the first stage, we use an exact hybrid model.
- $R_{i2} = R_{L1} = (1+h_{fe})R_E$
- Very high current gain.
- Unity voltage gain.

Example 1:

Determine the overall transconductance of the amplifier shown.



Previous Years' Question

In a multi-stage RC-Coupled Amplifier, the coupling capacitor:

a) limits the low frequency response
b) limits the high frequency response
c) does not affect the frequency response
d) blocks the d.c. components without effecting the frequency response

Sol:

$$I_0 = I_{C1} + I_{C2}$$

$$g_m = \frac{\partial I_0}{\partial V_{in}} = \frac{\partial (I_{C1} + I_{C2})}{\partial V_{in}} = \frac{I_{C1} + I_{C2}}{2V_{BE}}$$

$$= \frac{I_{C1} / V_{BE} + I_{C2}/V_{BE}}{2}$$

$$= \frac{g_{m1} + g_{m2}}{2}$$

Transconductance: $g_m = I_{CQ}/V_T$

$g_m = \frac{\partial I_C}{\partial V_{BE}}$: from transfer characteristics

Rack Your Brain

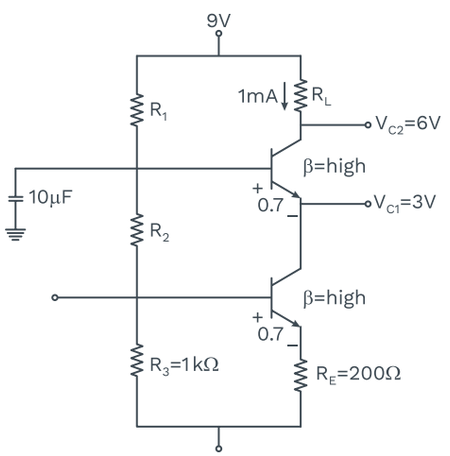
What is the role of internal capacitances in the frequency response of amplifiers?

Example 2:

The cascade amplifier circuit is shown in the figure, determine the value of R_1, R_2 and R_L, such that quiescent current through the transistor is 1mA, and collector voltages are V_{C1} = 3V and V_{C2} = 6V. Take V_{BE} = 0.7V and assume β to be high to neglect base current.

Grey Matter Alert!

Darlington pair amplifier provides very high current gain and input impedance.



Previous Years' Question

The Bandwidth of an n-stage tuned amplifier, with each stage having a bandwidth of B, is given by:

a) B/n **b)** $B/\sqrt{n}$

c) $B\sqrt{2^{1/n} - 1}$ **d)** $B/\sqrt{2^{1/n} - 1}$

Sol:

By KCL, we get,

$$R_L = \frac{9 - 6}{1mA} = 3k\Omega$$

$V_{B1} = 0.2 + 0.7$

$V_{B1} = 0.9V$

Emitter current through Q_2 = 1mA
(β is high)

The current throw registance R_3 is

$$I = \frac{0.9}{1K} = 0.9\,mA$$

$V_{R2} = 3.7 - 0.9 = 2.8V$

$$R_2 = \frac{2.8}{0.9} = 3.11k\Omega$$

$$R_1 = \frac{9 - 3.7}{0.9}$$

$R_1 = 5.89k\Omega$

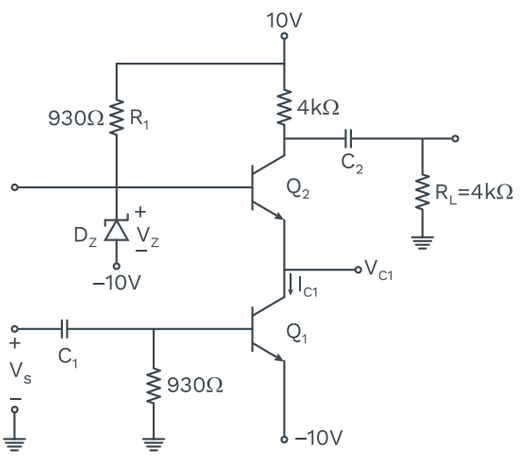
Previous Years' Question

From the measurement of the rise time of the output pulse of an Amplifier whose input is a small amplitude square wave, one can estimate the following parameter of the amplifier.

a) gain-bandwidth product
b) slew rate
c) upper 3-dB frequency
d) lower 3-dB frequency

Example 3:

The figure shows a two stage amplifier transistor Q_1 and Q_2 with currents gain β = 100 and also $\beta_{ac} = \beta_{dc} = \beta$. The zener diode has a breakdown voltage of 10.7V. Assuming D_z is in the breakdown region and if zener dynamic resistance is zero. The capacitance of C_1 and C_2 are large and provide zero impedance at the signal frequency.



1) Identity configuration of each transistor.
2) Determine quiescent quantifies I_{C1} and V_{C1}.

Sol:

$V_Z = 10.7V$

$\beta = 100$

$$\left.\begin{matrix} Q_1 = CE \\ Q_2 = CB \end{matrix}\right\} \text{Cascode}$$

DC Analysis,

$$I_{B1} = \frac{9 \cdot 3}{930} = 0.01A$$

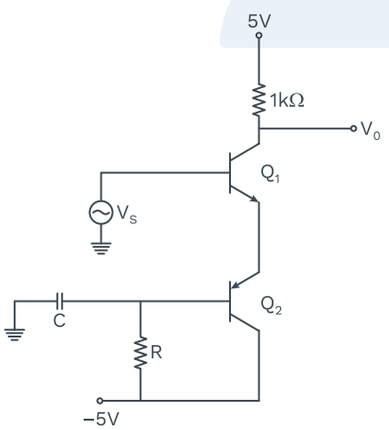
$$I_{C1} = \beta I_{B1} = 1A$$

$$V_{C1} = -10 + 10.7 - 0.7$$

$$V_{C1} = 0V$$

Example 4:

In the circuit shown Q_1 and Q_2 are biased at collector current of 2.6mA. Assuming that the transistor current gain is sufficiently high to assume collector current = emitter current and V_T = 26mV. Find the magnitude of voltage gain V_0/V_S in the mid band frequency range.



Sol:

$$g_m = \frac{I_C}{V_T} = 0.1$$

$$r_\pi \to \infty \qquad (\beta = \text{high})$$

$$g_m V_{\pi 1} = -g_m V_{\pi 2}$$

Previous Years' Question

A multistage Amplifier has a low-pass response with three real poles at s = $-\omega_1$, $-\omega_2$ and ω_3. The approximate overall bandwidth B of the Amplifier will be given by:

a) $B = \omega_1 + \omega_2 + \omega_3$

b) $\frac{1}{B} = \frac{1}{\omega_1} + \frac{1}{\omega_2} + \frac{1}{\omega_3}$

c) $B = (\omega_1 + \omega_2 + \omega_3)^{1/3}$

d) $B = \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2}$

$$V_{\pi 2} = -V_{\pi 1}$$

$$V_S = V_{\pi 1} - V_{\pi 2}$$

$$V_S = 2V_{\pi 1}$$

$$V_{\pi 1} = V_S/2$$

$$V_0 = -g_m V_{\pi 1} \times 1000$$

$$V_0 = -g_m \frac{V_s}{2} \times 1000$$

$$\frac{V_0}{V_s} = -\frac{0.1 \times 1000}{2} = -50$$

Frequency response of MOSFET:

- At low frequencies, the capacitances that dominate include coupling capacitor and bypass capacitor.
- At high frequencies, we consider the parasitic capacitances such as C_{gsp} and C_{gdp}, which are due to overlap of an oxide layer with drain and source region, also the internal capacitances such as C_{gs}, C_{ds}, C_{gd} dominate at high frequencies.

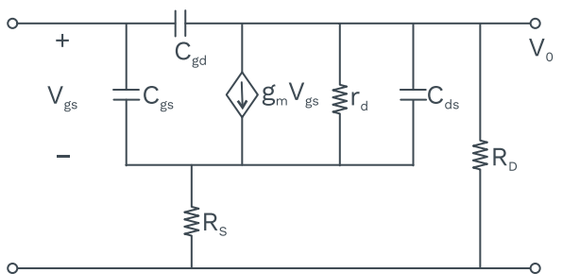


Fig. 6.27 Equivalent Circuit of MOSFET

These capacitances are considered at high frequency and treated as OC at low frequency.

Grey Matter Alert!

Lowest cut off frequency is the maximum of all the lower cut off frequencies and highest cut off frequency is the minimum of all higher cut off frequencies.

Previous Years' Question

An npn transistor (with C_μ = 0.3pF) has unity gain cut-off frequency f_T of 400 MHz at a DC-bias current I_C=1mA. The value of its C_π(in pF) is approximately (V_T = 26mV).

a) 15 **b)** 30
c) 50 **d)** 96

1) Unity gain bandwidth:

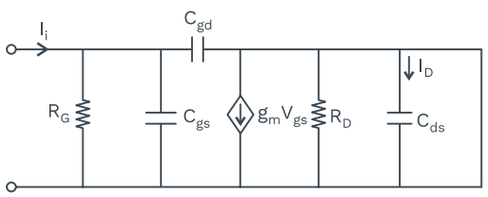


Fig. 6.28 Unity Gain Bandwidth

- We wish to determine unity gain BW by use of short circuit current gain.
- At high frequency, coupling and bypass capacitors have been shorted

$\therefore$ R_s is also shorted

R_G is very high, so input current is mainly dominated by capacitor current.

$$I_i = j\omega(C_{gs} + C_{gd})V_{gs}$$

By KCL, $I_0 = j\omega C_{gd}V_{gs} - g_m V_{gs}$

$$\text{Short circuit current gain} = \frac{I_o}{I_i}$$

$$= \frac{j\omega C_{gd} - g_m}{j\omega(C_{gd} + C_{gs})}$$

Practically, $\omega C_{gd} << g_m$

$$A_I = \frac{-g_m}{j\omega(C_{gd} + C_{gs})}$$

For unity gain $|A_I| = 1$

$$\frac{g_m}{\omega_T(C_{gs} + C_{gd})} = 1$$

$$\omega_T = \frac{g_m}{C}$$

$$f_T = \frac{g_m}{2\pi C}$$

Where, $C = C_{gs} + C_{gd}$

2) High frequency response:

The gain is constant over a wide range of frequencies known as mid-band gain, and at high frequencies, the gain reduces due to internal capacitances.

Previous Years' Question

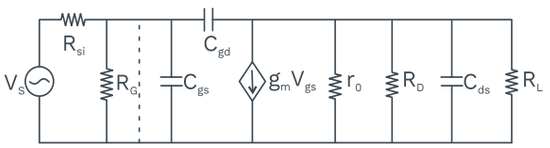
An npn BJT has g_m = 38 mA/V, $C_\mu = 10^{-14}$ F, $C_\pi = 4\times10^{-13}$ F and DC current gain β = 90. For this transistor f_T and f_β are:

a) $f_T = 1.64\times10^8$ Hz and $f_\beta = 1.47\times10^{10}$ Hz

b) $f_T = 1.47\times10^{10}$ Hz and $f_\beta = 1.64\times10^8$ Hz

c) $f_T = 1.33\times10^{12}$ Hz and $f_\beta = 1.47\times10^{10}$ Hz

d) $f_T = 1.47\times10^{10}$ Hz and $f_\beta = 1.33\times10^{12}$ Hz



Thevenin's equivalent circuit is,

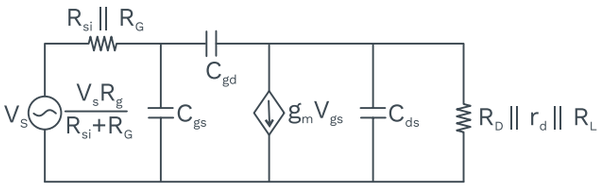


Fig. 6.29 High Frequency Response

By the use of Miller's theorem, C_{gd} can be resolved into two capacitances one at the input and the other at the output.

Input capacitance: $C_i = C_{gs} + C_{gd}(1–A_V)$

Output capacitance: $C_o = C_{ds} + C_{gd}(1–1/A_V)$

To determine the cutoff frequency, we have to determine resistance across capacitances.

$$f_{H,IN} = \frac{1}{2\pi(R_{si} \| R_G)C_{IN}}$$

$$f_{H,OUT} = \frac{1}{2\pi R_L' C_{out}} \quad ; R_L' = R_d \| r_d \| R_L$$

Overall cutoff frequency

$$f_H = \min[f_{H,IN}, f_{H,OUT}]$$

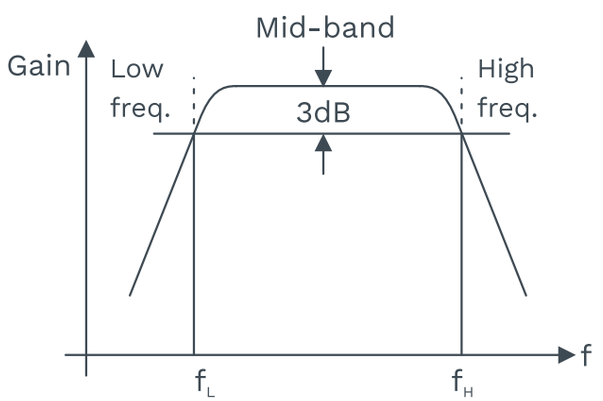


Fig. 6.30 Overall Frequency Response of MOSFET

3) Low frequency response:

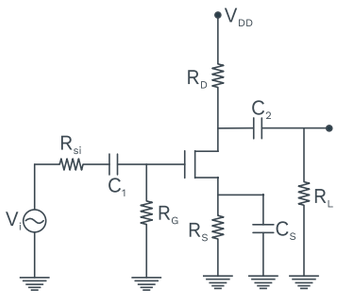


Fig. 6.31 LPF

- At low frequency we need to consider coupling capacitor, C_1 & C_2

Bypass capacitor, C_S

Frequency Range	Coupling & Bypass	C_{gd}, C_{ds}, C_{gs}
Low frequency	Consider	Open circuit (OC)
Mid band	Short circuit (SC)	Open circuit (OC)
High frequency	Short circuit (SC)	Consider

Table 6.2

To determine a lower cutoff frequency, we need to determine the resistance across each of these capacitances.

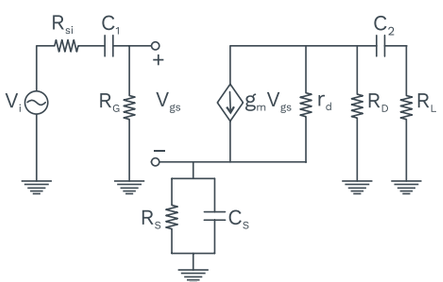


Fig. 6.32 MOSFET Equivalent Low Frequency Circuit

Across C_1: Short circuit, C_2 & C_S

R_{Th} across $C_1 = R_{Si} + R_G$

(short circuit voltage source)

$$f_{L1} = \frac{1}{2\pi(R_{Si} + R_G)C_1}$$

Across C_2: short circuit C_1 & C_S

R_{Th} across C_2: $(r_0 \parallel R_D + R_L)$

$$f_{L2} = \frac{1}{2\pi(r_0 \parallel R_D + R_L)C_2}$$

Across C_S: short circuit source

Short C_1 & C_2

Ignore r_0

$$f_{L3} = \frac{1}{2\pi\left(R_S \parallel \frac{1}{g_m}\right)C_S}$$

Overall lower cutoff frequency

$= f_L \max(f_{L1}, f_{L2}, f_{L3})$

MOS cascade:

If the first stage of the amplifier is a common source amplifier and the second state is a common gate configuration, then it is called as an MOS cascade connection.

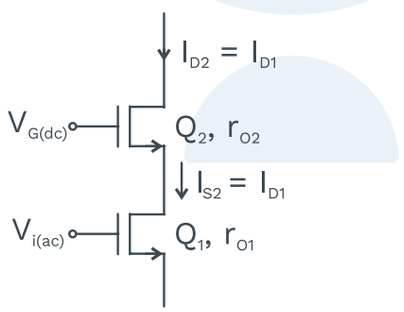


Fig. 6.33 MOS Cascade Amplifier

- Provides current buffering at the output of common source amplifying transistor.

  The input current of common gate transistor = output current of common gate transistor.

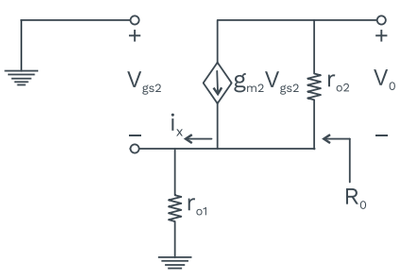


Fig. 6.34 MOS Cascade Amplifier Mode